TRANSPORTED ANTI-SPOOFING THRESHOLDS FAIL CONSERVATIVELY: MISSED SPOOFS AT A FALSE-ALARM RATE BELOW TARGET

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ABSTRACT

A speech-deepfake detector's threshold, set to 5% false-positive rate (FPR, bona fide rejected) on one channel, rarely holds it on another. Across 108 detector, source and target combinations built from the real transmissions of ASVspoof 2021 LA and four corpora, 77 realize an FPR more than $2\times$ off target (36 above, 41 below). Of the 72 cells within our own $\pm$5-percentage-point FPR tolerance, 41 fall below half the target: the tolerance reads clean where spoofs pass. In one AASIST example, a threshold calibrated on a PSTN transmission and deployed on the same recordings over G.722 realizes **0.42% FPR** while missing **70% of spoofs, against 0.66% at the deployment-calibrated threshold**. The failure persists on ASVspoof 5 when calibration and deployment share neither recordings nor speakers: 187 of 264 ordered pairs (71%) miss by more than $2\times$ across two detectors. A target-channel bona-fide order statistic restores *marginal expected* FPR control: at $N{=}500$, mean realized FPR is 4.91–5.06% over twelve detector–corpus cells, with mean FNR at most 0.5 points above a threshold fitted on all deployment bona fide in the run shown, in cells where that reference misses at most 50% of spoofs. The guarantee is marginal, not per deployment: its Beta dispersion law assumes iid sampling, and speaker clustering widens it on every tested cell. A label-free importance-weighted quantile does not reliably restore near-target FPR; a mixture-distance drift monitor fails under attack-prevalence shifts, and an entropy monitor fails its detection criterion under either drift definition. Threshold transport can therefore fail quietly, and target-channel calibration must report its sampling unit and dispersion.

***Index Terms*—** Anti-spoofing, audio deepfake detection, threshold calibration, conformal prediction, domain shift

1. INTRODUCTION

Audio anti-spoofing studies usually report performance at the EER threshold, an oracle operating point selected with labels from both evaluation classes. A deployed service has no such oracle: it must choose a threshold in advance and accept the resulting false-positive rate. A threshold set to 5% FPR on ASVspoof 2019 LA dev [1] realizes 22–98.5% FPR across 21LA/21DF [2], In-the-Wild [3], and BRSpeech-DF [4] for a strong open-source detector [5]; for XLS-R+SLS [6], it reaches 99.5%. Transfer failures of this kind are documented in [7, 8], and presentation can matter more than model scale [9]. LRLspoof [10] transfers an EER threshold across 66 languages but, without target-domain bona fide, can report only spoof rejection; our regime assumes the converse resource, target bona fide without target spoof labels, so we evaluate the FPR side.

We study the deployment behavior of a standard statistical repair: given N labeled target-domain *bona-fide* recordings and no spoof labels, the $\lfloor (N{+}1)\alpha \rfloor$-th order statistic has marginal rank validity under exchangeability without ties, and under iid continuous sampling its conditional FPR follows an exact finite-sample Beta law [11, 12]; conservative Neyman–Pearson variants give high-probability upper bounds [11]. Neither the statistic nor bona-fide-only audio calibration is new [13]. Our contribution is empirical: we measure transport across channels and corpora, its spoof-side cost, its persistence on disjoint data, and what alternative policies buy at the same labeling budget.

We make three contributions (details in §§4–5). (1) The measurement covers 108 deployment cells: two detectors $\times$ (42 ordered pairs of seven 21LA channel conditions + 12 ordered pairs of four corpora), reporting both the loss of false-alarm control and the spoof-side cost relative to each deployment's oracle. (2) We re-audit the bona-fide quantile against parametric and unlabeled alternatives at matched resource. (3) We report negative results for a label-free heuristic and two unlabeled drift monitors. We claim neither the threshold rule nor the practice of auditing both error rates at a fixed operating point [11, 12, 14, 15]. Our two detectors differ in both front-end and static accuracy, so $n{=}2$ cannot separate why they lose control by different amounts.

2. RELATED WORK

Order-statistic calibration is established: Neyman–Pearson methods control type-I error through class-0 order statistics [11, 16] and negative-only conformal calibration provides marginal guarantees [12]. In audio, CA-SOADD calibrates a bona-fide-only α-quantile [13], Reality Check fixes thresholds before evaluation [17], [7] audits threshold transfer and seven unlabeled corrections, and [8] documents bona-fide-driven cross-testing errors. Relative to that audit (one detector, two target corpora), we add a 108-cell fixed-FPR map with real transmissions, a finite-N target-bona-fide comparison, and a 264-pair disjoint replication. Auditing both error rates at a preset threshold is standard biometric practice [18, 14, 15], and [19] holds a cost-optimal threshold fixed across the same 21LA codecs but reports relative cost only; what remains open is how a threshold set on one channel behaves on another when only bona fide is available to reset it, which is what we measure. Threshold transfer is also examined for partial deepfakes [20], under bona-fide resource shifts [21], and in an industry report without reproducible figures [22]; our question is finite-sample reset from target bona fide alone. ReplayDF [23] reports channel-induced spoof-side degradation with stable bona-fide acceptance; we measure it at a fixed bona-fide error target.

Challenges audit fixed-cost operating points through t-DCF/a-DCF [14]; here spoof labels are unavailable for threshold selection but used for retrospective evaluation, and our standalone setting evaluates no tandem ASV system. Threshold calibration is distinct from the probability calibration evaluated with ECE in [24].

Prior work detects deployment drift and then retrains the model [25]; we hold the detector fixed and ask whether its threshold remains controlled. A pre-specified contamination test, replacing 25 of the 500 cohort recordings with spoofs, pushes realized FPR below target on seven of eight cells (0.31–4.96%); the eighth, overlap-dominated cell stays at 5.21%. We test only the score-weighting heuristic of §4, so its negative result is not attributed to covariate-shift weighting [26] or adaptive conformal inference [27].

3. THRESHOLD POLICIES

We treat spoof as the positive class. A detector emits score s (higher = bona fide), and deployment flags spoof when $s < t$, so $\mathrm{FPR}(t) = \Pr(s < t \mid \text{bona fide})$ is the bona-fide rejection rate ($P^{\mathrm{cm}}_{\mathrm{miss}}$ in the ASVspoof convention) and $\mathrm{FNR}(t) = \Pr(s \geq t \mid \text{spoof})$ the spoof acceptance rate. "Calibration" below means selecting t for a target FPR, not probability calibration. We compare policies by resource class. With **no target data**, the source dev threshold t at the target FPR is transferred unchanged. With **unlabeled target data**, we fit variants of the corrections of [7]: each transform is fitted to the unlabeled source and target mixtures separately, the 5% quantile of transformed source bona fide is taken, and that threshold is applied to transformed target scores. C1 is z-norm; C2 pseudo-labels the top and bottom score quartiles and fits a logistic transform by 50 Newton steps from w=1, b=0 with $10^{-6}I$ on the Hessian; C5 is AS-norm over the 100 nearest of at most 5,000 cohort embeddings, without self-exclusion. With N **labeled target bona-fide recordings**, we test three policies: (a) cohort z-norm standardizes scores by the cohort mean and standard deviation, then applies the source threshold in standardized units; (b) the Gaussian parametric quantile sets $t = \bar{s}_N - 1.645\,\hat{\sigma}_N$; and (c) the *conformal quantile* sets t to the k-th smallest cohort score, where $k = \lfloor (N{+}1)\alpha \rfloor$. Exchangeability without ties gives the last policy's marginal rank guarantee. Under iid continuous sampling from a fixed score distribution, its conditional FPR across calibration draws additionally follows $\mathrm{Beta}(k, N{+}1{-}k)$ [11, 12]. The **target-bona-fide oracle** sets the threshold realizing α on deployment bona fide; spoof labels measure its FNR but do not select it. The labeled policies use N=500 and B=1000, but Table 1 combines separate Monte Carlo runs: each baseline was paired with a quantile reference within its own run, and the displayed rows are unpaired means. **Definitions.** A detector–corpus cell is *overlap-dominated* when its oracle FNR at the target FPR exceeds 50%: it has no usable operating point under our joint criterion, is italicised in Table 1, and is excluded from claims about usable cells. The $\mathrm{Beta}(k, N{+}1{-}k)$ dispersion is an iid-continuous population reference, not the exact law of finite-pool draws without replacement; we report the held-out empirical spread separately over B=1000 draws. C4 (CORAL) of [7] additionally requires applying the frozen classifier head to transformed embeddings and is not evaluated.

4. EXPERIMENTS

Setup. We evaluate AASIST [28], SSL-AASIST [5], and XLS-R+SLS [6]. The first two use released 19LA-trained checkpoints; where official XLS-R+SLS scores are available, our scores agree at per-trial r=0.994. The corpora are ASVspoof 2021 LA and DF [2], In-the-Wild [3], BRSpeech-DF [4], and ASVspoof 5 eval [15] for replication. The 2021 keys carry *eval*, *progress* and *hidden* phases; the hidden phase removes non-speech by voice-activity detection and was not used in challenge results [29]. Non-speech is a documented shortcut for these detectors [19, 30], so the trimmed subset is a shortcut-removed benchmark [31] rather than a deployment channel, and mixing it into a calibration cohort mixes two score populations. A first version of this study did so; as a post-hoc protocol correction, every result below excludes it and keeps the two untrimmed phases, and restricting to the eval phase alone changes no count reported below. Each 21LA condition is then the same 2,356 bona-fide recordings of 67 speakers and 21,164 spoofs: six conditions are real transmissions, five through an Asterisk PBX over VoIP and one over a PSTN carrier, and one is the untransmitted source [2]. By contrast, 21DF is compression-only and untransmitted. Every detector is evaluated on the full 21LA and 21DF sets (16,492 and 20,637 bona fide; five 21DF spoof files were undecodable for our two detectors and excluded). ASVspoof 5 has no hidden phase. For cross-corpus transport the 21LA node is its untransmitted condition only (2,356 bona-fide and 21,164 spoof trials); Table 1 instead pools all seven conditions. All thresholds target α=5%. Labeled policies use the same paired cohort draws (B=1000), and FPR is measured on held-out data.

FPR control and FNR cost. At N=500, the conformal quantile's mean realized FPR is 4.91–5.06% across the 12 detector–corpus cells (Table 1), and in a separate 1000-draw diagnostic on the same score sets the 2.5–97.5 percentile spread of realized FPR lies within 2.8–7.6% on every cell, close to the iid Beta reference. On every cell with a usable operating point, the quantile's FNR premium over the oracle threshold is $\leq$0.5 points.

Matched-resource policy comparison. The lower block of Table 1 compares the policies of §3 on SSL-AASIST at N=500 over 1000 draws. The conformal quantile realizes 4.9–5.0% FPR on all four corpora, as marginal rank validity predicts. Naive transfer and the unlabeled corrections miss the target by up to 95 pp. The Gaussian parametric quantile is the fair competitor at the same N because it also targets a target-domain FPR; it misses by up to 6.9 pp and collapses to 0% on BRSpeech. Cohort z-norm does not target a target-domain FPR, so its larger errors (6/8 detector–corpus cells beyond $\pm$2 pp, up to 50.8 pp for XLS-R+SLS) are a category difference rather than a direct contest.

Drift is detector-conditioned (Fig. 1). A cell reports the mean deployment FPR over 1000 thresholds, each set from 500 source bona-fide recordings. Its severity is the $\log_2$ ratio of $\max(\mathrm{FPR}, 3/n)$ to α, with n the deployment bona-fide count. Our $\pm$5-point band around a 5% target includes zero by construction, so it cannot distinguish a collapsed threshold from a controlled one; an absolute tolerance narrower than the target would not share this defect. Over the 108 cells, 77 miss target by more than 2$\times$ (57 of the 84 within-corpus cells). On the ratio scale, 41 of the 72 cells inside our own $\pm$5 pp tolerance on realized FPR miss target by over 2$\times$, necessarily conservatively, since that tolerance excludes a liberal 2$\times$ miss by construction. The 2$\times$ bar is post-hoc, and the count depends on that choice: it ranges from 54 of 72 at 1.5$\times$ to 29 of 72 at 4$\times$. We therefore attach no interval. 58 of those 72 cells are within-corpus, where calibration and deployment are the same 2,356 recordings from the same 67 speakers under seven 21LA channel conditions, so they are not independent observations and no significance claim is made of them; the replicate below tests whether the finding survives this dependence. Their spoof-side cost is measured against the oracle threshold for that deployment, not against the calibration condition, and reaches +0.69 (AASIST, PSTN$\to$G.722, whose 0.42% FPR is about 10 false alarms per draw among 2,356 bona fide; two cells score marginally higher but their FPR rests on fewer than five expected false alarms, $\mathrm{FPR} < 5/n$, and is censored). Every AASIST cell calibrated on PSTN pays at least +0.40; the same threshold misses 31% of PSTN spoofs, and PSTN is AASIST's least separable condition (AUC 0.956 against 0.975–0.997), so its threshold sits below the bona-fide scores of cleaner channels. Per calibration draw a positive spoof-side cost places the threshold below oracle and so inside the $\pm$5 pp tolerance; empirically every positive-price cell passes it, which is why the additive test cannot reveal this cost. AASIST pays the larger spoof-side cost. **Replication on recording-disjoint data.** Because the grid above reorders one 2,356-recording set, we repeat the experiment on ASVspoof 5 eval [15], where calibration and deployment share neither recording nor speaker. The grid contains 11 organizer-applied codecs plus the unprocessed source and 737 speakers. It is restricted to the 80.5% of bona-fide sources appearing under one condition and split 50/50 by speaker. **Over both detectors the transported threshold misses target by over 2$\times$ on 187 of 264 pairs (71%), against 68% on the grid above**: the effect persists without shared recordings or speakers; the numerical rate difference is confounded by corpus, codecs and attacks. **The spoof-side replication is narrower.** A no-codec pilot with 1,500 recordings per class gave oracle FNRs of 63.9% (AASIST) and 7.9% (SSL-AASIST); in the disjoint evaluation every AASIST destination and six of the twelve SSL-AASIST destinations exceed 50% oracle FNR. Spoof-side conclusions are therefore restricted to SSL-AASIST's other six destinations (66 ordered pairs), where 49 miss the FPR target conservatively by more than 2$\times$ and 51 pay a positive spoof-side cost. The flagship AASIST cell does not replicate; the spoof-side axis here rests on one detector. On this deployment half (34,539 bona fide, 312,440 spoofs) the pooled EER is 30.6% for AASIST and 11.8% for SSL-AASIST, not a full-protocol value. On three further detectors with released 21LA scores (XLS-R+SLS, XLSR-Mamba [32], and the fixed-size XLSR-Conformer [33]), the transported threshold misses target by more than 2$\times$ on 25–33 of the 42 channel pairs each (88 of 126), while their spoof-side cost on PSTN$\to$G.722 stays below 3 points: the FPR-axis failure is detector-general, the spoof-side magnitude is not.

Monitoring. We evaluated two unlabeled drift monitors, pre-specified with the acceptance criterion TPR $\geq$ 0.8 and FPR $\leq$ 0.2 for flagging drifted cells. The flag definition was changed after results existed, from "realized FPR more than 2 pp off target" to $|\log_2(\mathrm{FPR}/\alpha)| > 1$; under the pre-specified definition no monitor has an operating point. Under the post-hoc alternative definition the mixture-W_1 monitor, the W_1 distance between deployment and calibration mixture-score distributions, has an in-sample operating point on SSL-AASIST only (threshold selected on the same cells), but multiplying the deployment spoof-to-bona-fide odds by 0.5 and 1.5 with the bona-fide set fixed pushes 19/19 and 18/

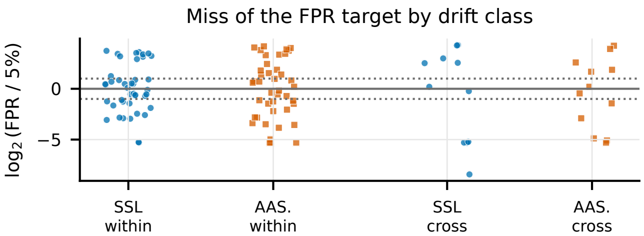

Fig. 1. Drift map: signed $\log_2(\mathrm{FPR}/5\%)$ per deployment cell (within-21LA channel pairs vs cross-corpus; colour/marker = detector). Under a post-hoc 2× two-sided bar, 41 of the 72 cells inside the ±5 pp tolerance are off target (54 at 1.5×, 29 at 4×; no interval, §4), every one conservative. Severity is $\log_2(\max(\mathrm{FPR}, 3/n)/5\%)$ (§4).

19 benign cells over it. The entropy monitor, the mean binary entropy of $p_i = \mathrm{clip}(\sigma((s_i - \mu)/\varsigma), 10^{-6}, 1 - 10^{-6})$ over deployment scores with μ, ς the calibration-mixture mean and standard deviation, alerting on large values, fails on both detectors under either definition. **Label-free heuristic.** We test an importance-weighted quantile that reweights the 500 labeled source-domain bona-fide scores of the calibration cohort by a clipped ratio of unlabeled deployment to calibration *mixture-score* densities (15 equal-width bins over the pooled score range, ratio clipped to $[0.1, 10]$, 200 draws); it uses no target labels. It places 17/108 cells within 2 pp of the FPR target, compared with 20/108 unweighted, so it does not reliably restore near-target FPR across this grid; we did not evaluate its spoof-side error. It is not the construction of [26], which needs a target/source bona-fide covariate ratio and a test-point weight; the negative result is limited to the heuristic. An adaptive-conformal FPR update [27] needs labeled bona-fide outcome feedback; labels on flagged items alone, or the unlabeled mixture flag rate, do not identify FPR when attack prevalence and the spoof-side detection rate are unknown, and we evaluate no estimator that adds identifying assumptions.

5. DISCUSSION

Calibration and detection are distinct. Target-channel calibration controls the *marginal mean*, not every realized threshold. Under iid continuous sampling from a fixed distribution, a single N=100 draw has only 65.5% probability of landing within ±2 FPR points; at N=500, the probability is 96.2%, with a 3.26–7.06% central 95% interval. **Operational implications.** Report FNR at controlled FPR alongside EER, choose N from the dispersion required by the deployment, ship a designated bona-fide calibration split, and recalibrate after a channel change. **Limitations.** The Beta dispersion law assumes iid continuous recordings from a fixed deployment speaker mixture. A post-hoc speaker-disjoint diagnostic on 21LA, which calibrates on whole speakers until the cohort holds at least 500 recordings (median 15 speakers) and evaluates on the rest, widened the 5th–95th percentile spread of realized FPR on all six tested detector–condition cells (AASIST and SSL-AASIST on the untransmitted, PSTN and GSM conditions) to 6.0–10.1 points (means over ten seeds of 400 draws), against a median of 3.4–3.5 under a null that reassigns individual bona-fide scores to fixed speaker-sized groups (200 permutations); deployment to new speakers therefore requires speaker-level validation. Conclusions rest on three detectors (the drift map and replicate on two) and one language-shift corpus (n=1), and assume bona-fide labeling is cheap relative to spoof labeling, as when a deployment can verify its own recordings.

Table 1. Upper block: mean FNR (%) over B=1000 paired N=500 cohorts at the conformal quantile targeting FPR=5%; parentheses give mean FNR minus deployment-oracle FNR, in points. Per-condition 21LA EER ranges: 1.9–11.1 (AASIST), 0.2–1.0 (SSL-AASIST), 0.5–3.5 (XLS-R+SLS). Lower block: mean realized FPR of each policy of §3 for SSL-AASIST at the same N and B from separate runs (naive transfer and C2/C5 are deterministic); C5 reaches its FPR only at FNR ≈99%. Italics: overlap-dominated cells (oracle FNR >50%), where FPR control holds but no usable operating point exists. Code, score tables and results: `https://github.com/rvirgilli/speech-deepfake-threshold-transport/tree/a2-icassp2027-final`.

System	21LA	21DF	ITW	BRSpeech
AASIST	19.1 (+0.4)	36.7 (+0.1)	*95.2 (−0.1)*	*84.0 (−0.3)*
SSL-AASIST	0.2 (+0.0)	6.9 (+0.1)	18.3 (+0.2)	*98.5 (0.0)*
XLS-R+SLS†	1.5 (+0.1)	0.7 (+0.1)	11.3 (+0.5)	*91.5 (+0.4)*
EER (%) on the same trials				
AASIST	8.3	14.2	43.0	36.1
SSL-AASIST	0.8	5.9	11.2	34.7
XLS-R+SLS†	2.8	2.0	7.5	34.2
Realized FPR (%) per policy, SSL-AASIST, N=500				
Naive transfer	22.2	38.4	98.5	95.7
C2 temp./shift	0.3	1.4	100.0	100.0
C5 AS-norm	1.4	6.4	0.0	0.1
Cohort z-norm	7.4	20.3	11.9	23.2
Gaussian q.	5.6	11.9	5.7	0.0
Conformal q.	5.0	5.0	5.0	4.9

† Official author-released scores for 21LA, 21DF and ITW; BRSpeech scored by us with the released checkpoint.

6. CONCLUSION

Threshold transport is easy to mis-audit: conservative collapse makes FPR read below target while missed spoofs multiply. A target-channel bona-fide order statistic restores marginal expected control; its dispersion depends on the sampling unit, and it cannot repair class overlap.

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